Raw Materials Classification Using Image Texture

The goal of this project is to build an automated system for material classification using deep learning. Material recognition is an important task in computer vision, with applications in robotics, manufacturing, and scene understanding. The challenge lies in accurately identifying different materials (e.g., glass, metal, plastic, wood) under varying lighting, textures, and backgrounds.

Introduction

For this project, we used the MINC (Materials in Context) dataset, a well-established benchmark for material classification. The dataset provides a large number of labeled images across multiple material categories. Images were preprocessed and split into three subsets: training, validation, and testing, ensuring that the model can generalize well to unseen data.

Problem Definition

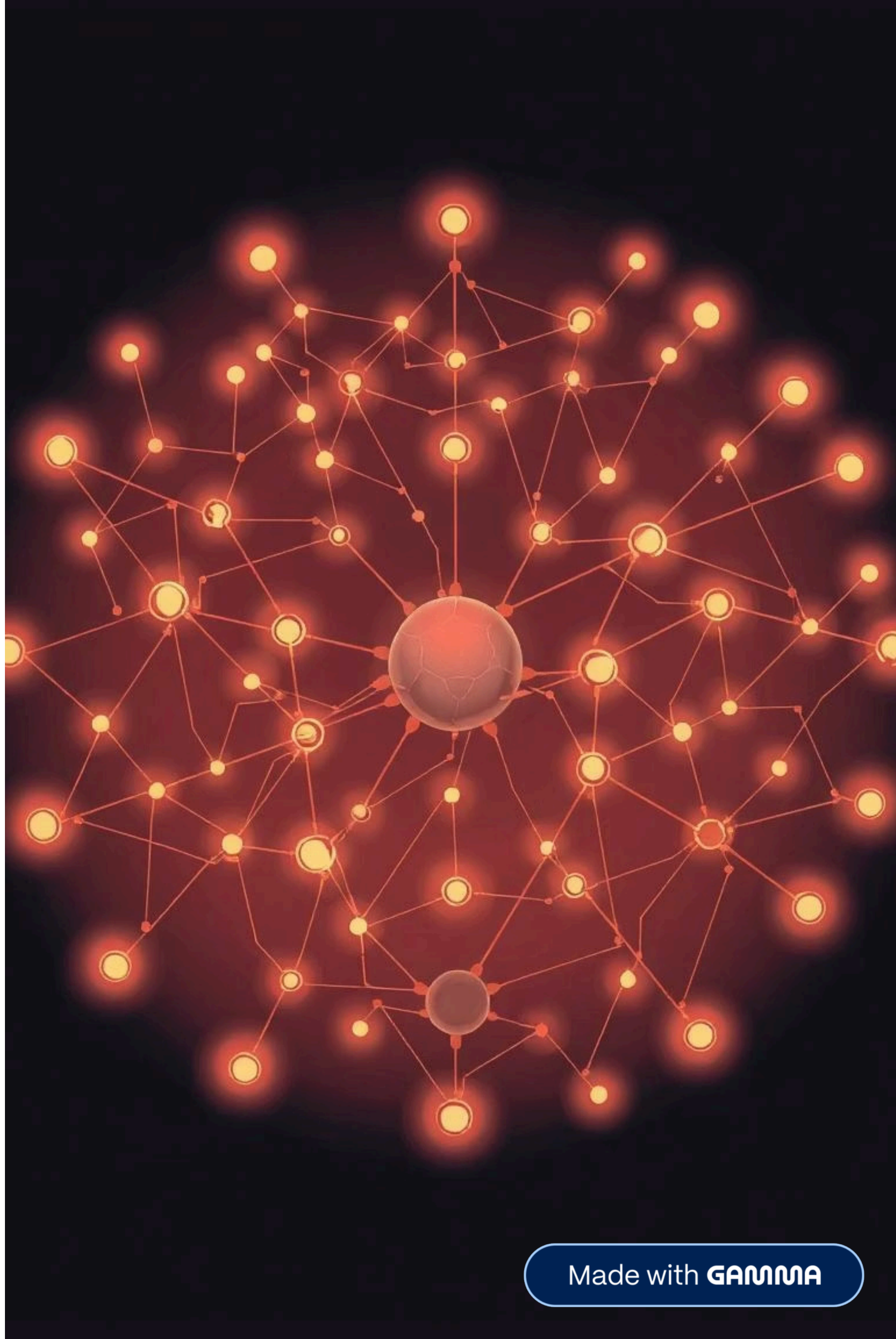
In real-world industrial environments, material classification systems face multiple challenges:

- Variations in lighting intensity and direction
- Shadows caused by machinery or material stacking
- Reflections and glare on metal and glass surfaces
- Similar visual textures between different materials

Traditional rule-based or color-based image processing techniques often fail under such conditions. Therefore, a robust, data-driven approach that can generalize across different visual scenarios is necessary.

Problem Statement

How can deep learning models accurately classify raw materials based on texture, even under challenging lighting and environmental conditions?



Dataset Selection

Source: mcimpoi/minc-2500_split_1 · Datasets at Hugging Face

• The dataset was obtained from MINC (Materials in Context), a well-known benchmark for material recognition.

Data Split:

- **Train set:** Used for model training
- **Validation set:** Used for monitoring performance during training and tuning hyperparameters
- **Test set:** Used for final evaluation

Data Augmentation (Training):

- Random horizontal flip
- Random rotation (±15 degrees)
- Color jitter (brightness, contrast, saturation, hue)

Normalization:

• All images were normalized using the mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225] to match the pretrained models' expectations.

Model Comparison and Results

Three deep learning architectures were used for material classification on the MINC dataset (classes: glass, metal, plastic, wood). All models employed transfer learning with pretrained weights, progressive unfreezing, and label smoothing to improve generalization.

EfficientNetB1

Accuracy: **86.91%**

ResNet101

Accuracy: **86.91%**

VGG16

Accuracy: 80.66%

EfficientNetB1 Performance

Overall Metrics:

- Accuracy: **86.91%**
- Precision: **87.53%**
- Recall: **87.04%**

Per-Class Metrics:

Class	Accuracy	Precision	Recall
Glass	0.8357	0.8550	0.8357
Metal	0.8306	0.8834	0.8306
Plastic	0.8995	0.8020	0.8995
Wood	0.9159	0.9608	0.9159

Observations:

- Wood is recognized with very high precision and recall.
- Plastic has high recall but slightly lower precision.
- EfficientNetB1 achieves the best balance between precision and recall across all classes.

ResNet101 and VGG16 Performance

ResNet101

Overall Metrics:

- Accuracy: **86.91%**
- Loss: 0.4311

Per-Class Metrics:

Class	Precision	Recall
Glass	0.8745	0.7746
Metal	0.8913	0.7492
Plastic	0.7487	0.9222
Wood	0.8766	0.9159

Observations:

- Plastic has very high recall but lower precision, indicating some misclassifications.
- Glass and Metal have high precision but moderate recall.
- ResNet101 performs slightly worse than EfficientNetB1 in balancing precision and recall across classes.

VGG16

Overall Metrics:

- Accuracy: **80.66%**
- Loss: 0.7102

Per-Class Metrics:

Class	Precision	Recall
Glass	0.8485	0.7037
Metal	0.7991	0.6213
Plastic	0.6341	0.9157
Wood	0.8403	0.7980

Observations:

- Plastic has very high recall but low precision, leading to many false positives.
- Glass and Metal show moderate recall and confusion between classes.
- VGG16 is weaker overall compared to ResNet101 and EfficientNetB1.

Summary & Insights

Model	Accuracy	Notes
EfficientNetB1	86.91%	Best balance between precision & recall; robust to class imbalance.
ResNet101	86.91%	Strong model, high precision for most classes, slightly less balanced.
VGG16	80.66%	Lower overall accuracy; high recall for plastic but suffers from confusion.

Data Preprocessing and Augmentation

Image Preprocessing

- All images resized to **224 × 224 pixels**
- Pixel values normalized using ImageNet mean and standard deviation
- Labels encoded into categorical format

Data Augmentation

To simulate real factory conditions and improve robustness, extensive augmentation techniques were applied:

- Random brightness and contrast adjustment
- Random gamma correction
- Random rotations and zooming
- Horizontal and vertical flipping
- Shadow simulation

These techniques force the model to focus on intrinsic texture features rather than lighting or color cues.

Challenges and Mitigation

Class Imbalance

- Some classes (e.g., wood) had fewer samples compared to others (e.g., wood), which made the models biased toward the majority classes.
- Weighted sampling and careful loss functions (label smoothing) were used to mitigate this issue.

Small Dataset Size

- Even with transfer learning, the limited number of images per class made generalization challenging.

Overfitting

- High-capacity models tended to overfit, especially on classes with fewer samples.
- Data augmentation, dropout, and label smoothing were applied to counter overfitting.

Low Accuracy on Certain Classes

- Some classes, such as plastic or metal, showed lower precision or recall compared to others due to visual similarity and texture variations.

Mitigation:

- Class weights were used in the loss function to give more importance to underrepresented classes.
- WeightedRandomSampler ensured balanced sampling during training.
- Data augmentation focused on simulating real-world variations (lighting, shadows, reflections) to improve generalization.
- Progressive unfreezing allowed fine-tuning deeper layers carefully to capture subtle texture differences without overfitting.